INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

# INDICATOR, POSITION, TAKEOFF TRIM ALU-12/A

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the requirements for a 1-inch takeoff trim position indicator, designated ALU-12/A.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | Boxes Wood, Cleated Plywood |
| PPP-B-636 | Box, Shipping, Fiberboard |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-T-152 | Treatment, Moisture- and Fungus-Resistant, Of Communications, Electronic, And Associated Electrical Equipment |
| MIL-S-5002 | Surface Treatments And Metallic Coatings For Metal Surfaces Of Weapons Systems |
| MIL-I-7057 | Indicators, Synchro, Aircraft, General Specification For |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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| | |
|---|---|
| MIL-C-14806 | Coating, Reflection Reducing, For Instrument Cover Glasses And Lighting Wedges |
| MIL-L-27160 | Lighting, Instrument, Integral, White, General Specification For |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-100 | Engineering Drawing Practices |
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of US Military Property |
| MIL-STD-202 | Electronic And Electrical Component Parts, Test Methods |
| MIL-STD-454 | Electronic Equipment, Standard General Requirements |
| MIL-STD-461 | Control Of Electromagnetic Interference Emissions |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement Of |
| MIL-STD-704 | Aircraft Electric Power, Characteristics And Utilization Of |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-970 | Standards and Specifications, Order Of Preference For The Selection Of User |
| MIL-STD-1523 | Age Controls of Age-sensitive Elastomeric Material |
| MS3126 | Connector, Plug, Electric, Crimp-type, Straight, Bayonet Coupling |
| MS28042 | Clamp, Mounting, Aircraft, Instrument |
| MS33558 | Numerals And Letters, Aircraft Instrument Dial, Standard Form Of |
| MS33639 | Case, Instrument, Clamp-Mounted, Aircraft |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 1700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

| | |
|---|---|
| ASTM D3951 | Packaging, Commercial |

(Application for copies should be addressed to: ASTM, 1916 Race Street, Philadelphia, PA 19103.)
(Non-Government Standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## 3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials

3.3.1 Non magnetic materials. Non magnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Combustible materials. Materials that will support combustion, maintain flame, or cause fire shall not be used.

3.3.3 Toxic materials. Materials shall not liberate toxic or corrosive fumes or gases which would combine with the atmosphere to form an acid or corrosive alkali.

3.3.4 Metals. The use of inherently corrosive-resistant metals shall be used wherever possible. Metals that are not inherently corrosion resistant shall be treated in accordance with MIL-S-5002.

3.3.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. If necessary to use dissimilar metals, their use and protection shall be in accordance with MIL-STD-454, requirement 16.

3.3.5 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated in accordance with MIL-T-152.

3.3.6 Synthetic rubber materials. The use and processing of age-sensitive elastomeric materials shall be in accordance with MIL-STD-1523.

3.4 Design and construction. Except as specified herein, the indicator shall be designed and constructed in accordance with MIL-I-7057 for one synchro. The design shall be such that two indicators may be operated in parallel without degradation of the performance requirements of this specification. The indicator shall contain a synchro torque repeater for receiving and displaying horizontal tail pitch-trim information. The display shall conform to figure 1.

3.4.1 Synchro torque transmitter. The input signal shall be provided by a synchro torque transmitter having the following characteristics:

a. Input voltage: 26V 400 Hz

b. Input current: 110 ma

c. Input power: 0.69 watt

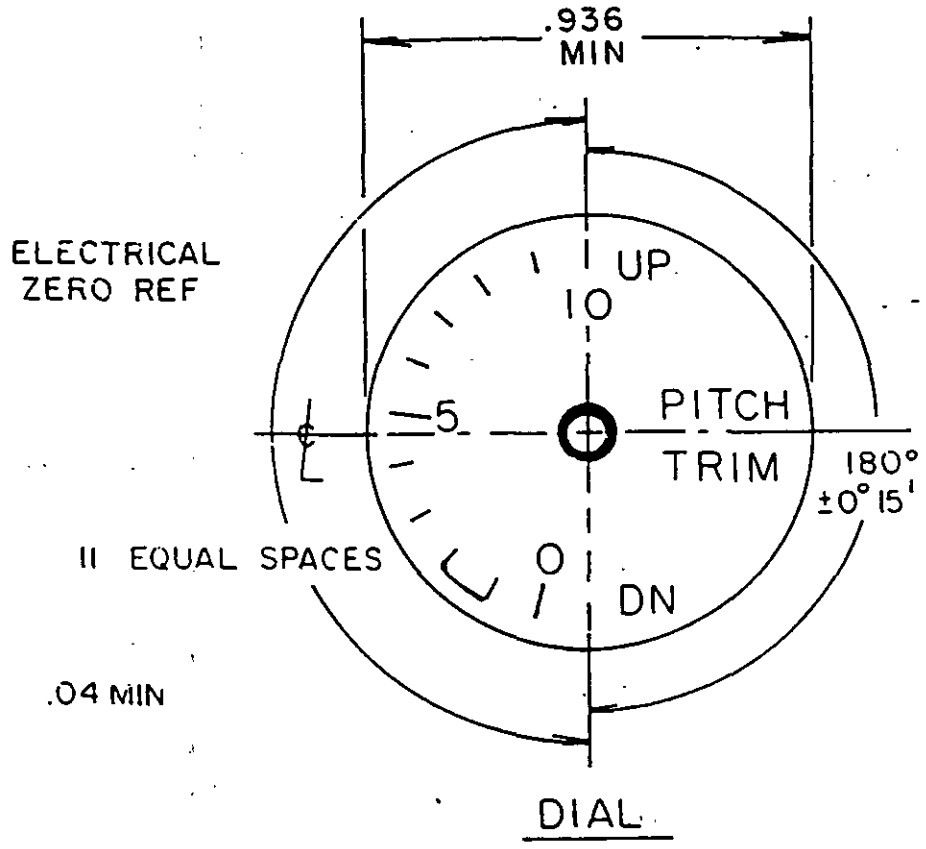
d. Output voltage: 11.8V

e. Sensitivity: 206 mv/deg

f. Phase shift: $10^{o}$



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Dial background Durable Black, Color No. 37038 per Fed-Std-595

FIGURE 1. Dial



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g. D-C resistance: Rotor 38 ohms; stator 12 ohms

h. Impedence: $Z_{RO}$ 58 + J234 ohms; $Z_{SO}$ 12.5 + J40 ohms

i. Maximum null voltage-millivolts: 30

j. Accuracy: Maximum error 7 minutes; error spread 14 minutes.

3.4.2 Electrical zero. Electrical zero shall be within 0.010 inch of the electrical zero reference mark on the dial. The electrical interface shall conform to figure 2.

3.4.3 Case. The case shall be nominal 1-inch size, shall conform to figure 3 and MS 33639, and shall be hermetically sealed. Hermetic sealing shall be so accomplished that the seal will not be affected by the environmental conditions specified herein. The exterior of the case shall be finished in durable dull black, color No. 37038 of FED-STD-595.

3.4.3.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. Other concentrations of helium shall not be used except with prior approval of the procuring activity. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.4.4 Electric power. The indicator shall operate from nominal 26V single-phase a-c power in accordance with MIL-STD-704 except the ranges shall be from 22 to 30V and 320 to 480 Hz. Power supplied to the lighting circuit shall be 0 to 5V ac or dc.

3.4.5 Electrical connector. An electrical connector shall be provided which shall conform to figure 3.

3.4.6 Lighting. The indicator shall be integrally lighted in accordance with MIL-L-27160. Integral lighting of the indicator shall be comparable in brightness, contrast, and color to the prime standard specified in 3.4.6.1 when the indicator is energized with 3.00 $\pm$0.01V and the prime standard is energized with 3.00 $\pm$0.16V.

3.4.6.1 Prime standard. A prime standard shall be provided which shall be either an indicator conforming to this specification or a lighted model. The prime standard lighting shall conform to MIL-L-27160. The design, color, quality of lighting, and calibration shall be subject to the approval of the procuring activity. The standard shall be calibrated every 50 hours of operation.

3.4.7 Dial

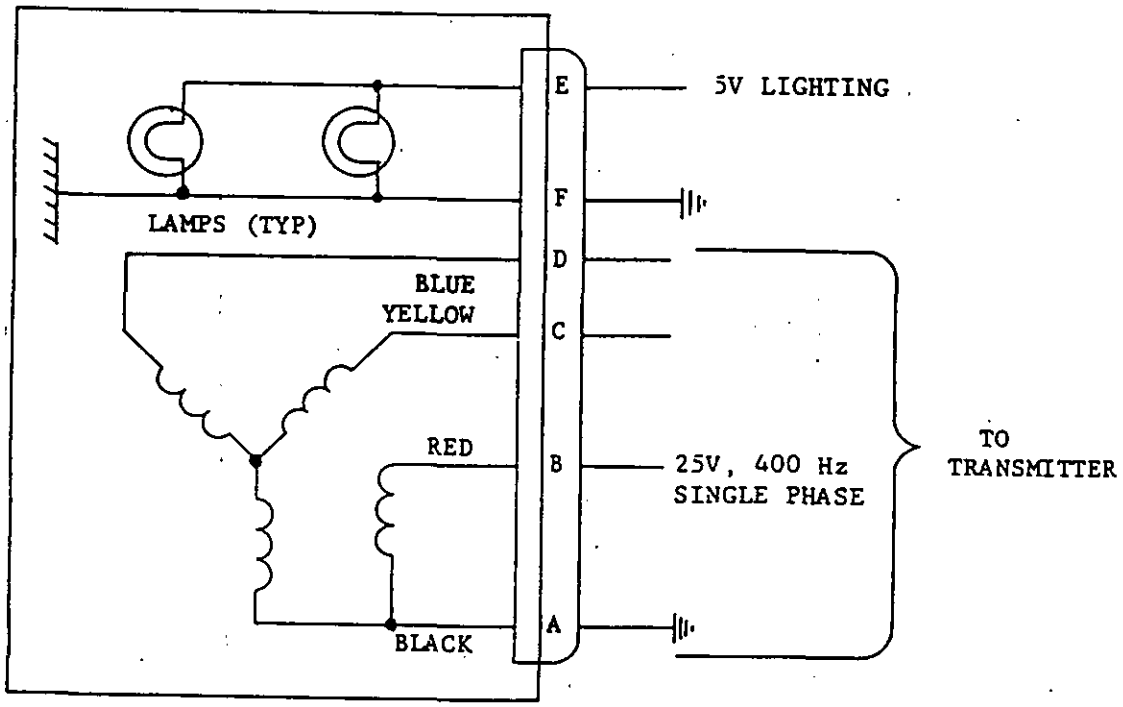
3.4.7.1 Letters, numerals, and markings. All letters, numerals, and markings shall be visible when viewed from any position within the frustum of a cone the side of which makes an angle of 30° with the normal to the center of the coverglass surface. Letters, numerals, and markings shall be in accordance with MS33558 and table I.

3.4.7.2 Dimensions and colors. Dimensions and colors shall conform to figure 1. The display shall be as close to the coverglass as practicable to minimize parallax.

3.4.8 Coverglass. The coverglass shall be installed from the inside of the case and shall be hermetically sealed within the case. It shall be flat and free from discoloration's, scratches, and striae that would interfere with reading of the pointer relative to the dial.



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NOTE: GROUND WIRE PIN F OF THE LIGHTING CIRCUIT SHALL BE INTERNALLY CONNECTED TO THE INSTRUMENT CASE.

FIGURE 2. Electrical interface



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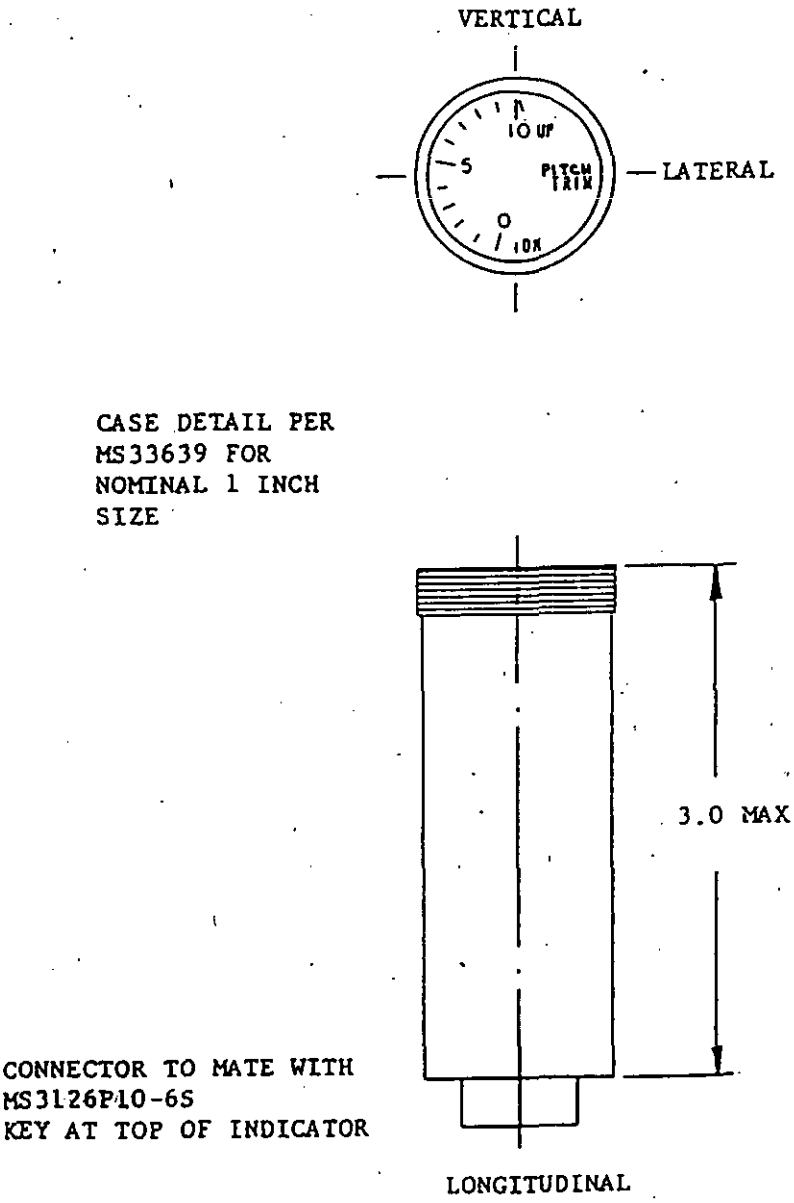



FIGURE 3. Outline dimensions



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3.4.8.1 Reflection-reducing coating. A reflection-reducing coating conforming to MIL-C-14806 shall be applied to each surface of the cover glass and lighting wedge.

Table I. Dial Marking Instructions

| Dial Markings | Height or Length Inches | Width of Line or Graduation Inches ±0.005 |
|---|---|---|
| Major graduations | 0.109 ±0.010 | 0.030 |
| Minor graduations | 0.063 ±0.010 | 0.020 |
| Graduations at Electro zero | 0.030 ±0.010 | 0.015 |
| Lettering PITCH, TRIM, UP, DN, 0, 5, & 10 | 0.094 ±0.016 | 0.020 |

3.4.9 External bonding. Provisions shall be made for electrically bonding the indicator case to the aircraft structure.

3.4.10 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,500 hours demonstrated to a 90 percent confidence level.

3.4.10.1 Longevity. The indicator shall have an operating life span (longevity) of not less than 3,750 hours before wearout failures occurs or the equipment consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity. _

3.4.11 Maintainability. No preventive maintenance shall be required throughout the service life of the indicator. Fault isolation and repair within the hermetically sealed indicator shall be accomplished at depot level without the use of special tools.

3.4.12 Lubrication. The lubrication used shall be such that the indicator will operate satisfactorily when subjected to the environmental conditions specified herein. Parts that are permanently sealed shall require no additional lubrication throughout their operational life.

3.5 Performance. The indicator shall operate as specified herein during and after subjection to the following conditions:

a. Temperature: Operating temperatures ranging from -54° to +71° C and non operating temperatures ranging from -62° to +71° C.

b. Salt fog: Salt sea atmosphere for 48 hours.

c. Shock up to 20g for 11 milliseconds.

d. Vibration: Withstand vibration as shown on figure 4, curve A, and operate within specification limits when subjected to vibration as shown on figure 4, curve B.



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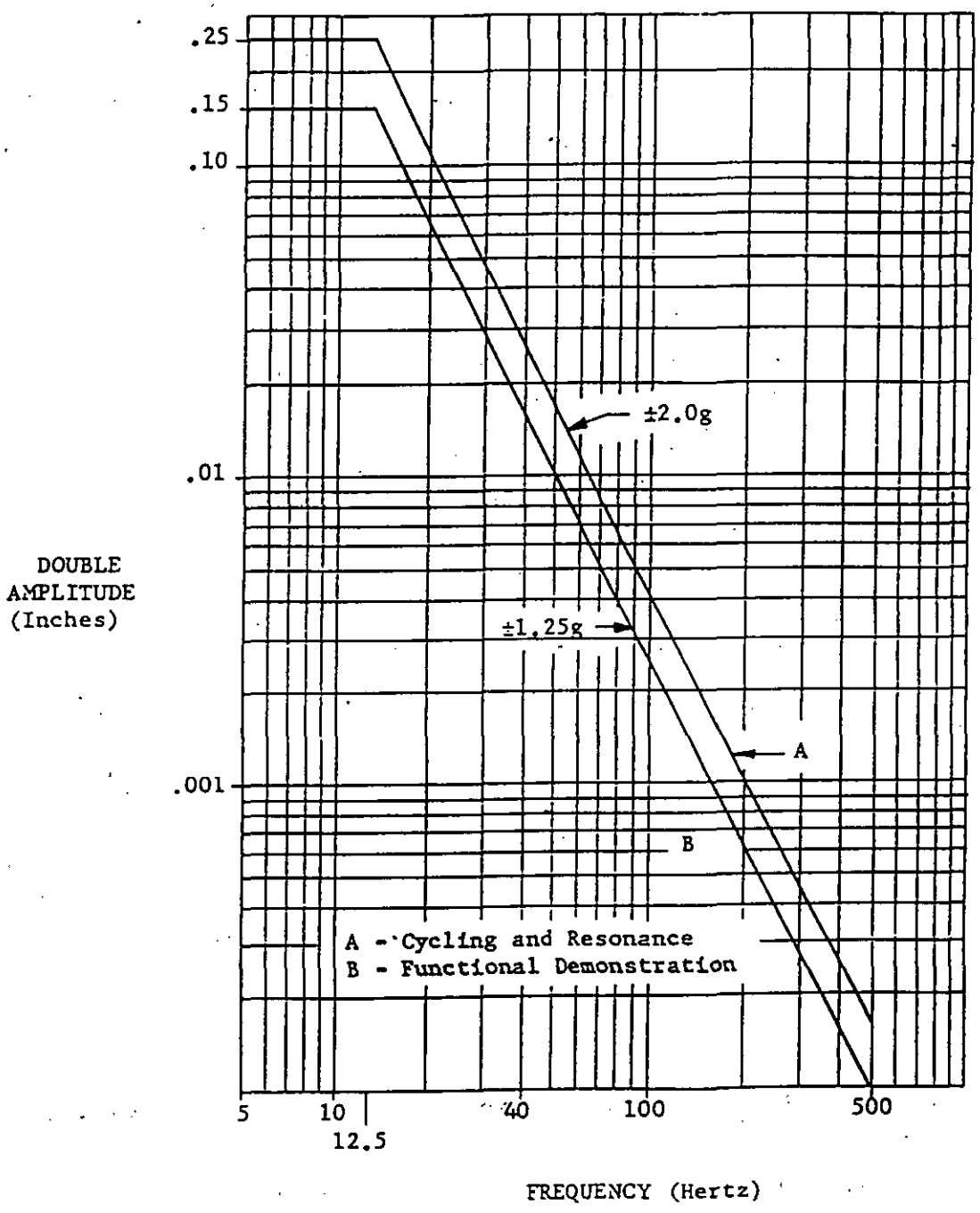



FIGURE 4. Vibration test amplitudes



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e. Attitude: Between $0^{o}$ and $360^{o}$ in pitch and roll as encountered during flight.

3.5.1 Scale error. Scale error for a single indicator shall not exceed $\pm$ 2 angular degrees at room temperature, $+71^{o}$, and $-54^{o}$ C. When two indicators are connected in paralleled with a single transmitter, the scale error shall not exceed 1.25 times the scale error for a single indicator.

3.5.2 Friction error. Friction error shall not exceed $\pm$2 angular degrees at room temperature and + $71^{o}$ C ; and shall not exceed $\pm$ 4 angular degrees at - $54^{o}$ C.

3.5.3 Position error. Position error shall not exceed $\pm$2 angular degrees.

3.5.4 Vibration error. Pointer oscillation and variation shall not exceed $\pm$2 angular degrees.

3.5.5 Case leakage. The indicator shall be so sealed that the maximum leak rate at a pressure differential of approximately 1 atmosphere will not permit more than ten percent loss of the total filling medium after 1,000 hours.

3.5.6 Fogging. There shall be no evidence of moisture or oil fog inside the coverglass when the indicator is energized, placed in a $71^{o}$ $\pm 2^{o}$ C controlled ambient temperature for a minimum of one hour, and an ice cube then rubbed on the coverglass for one to two minutes while the indicator is still at the specified temperature.

3.5.7 Insulation resistance. The insulation resistance between all external connections of each mutually insulated circuit and between each external connection and the case shall be a minimum of 100 megohms.

3.5.8 Dielectric strength. The dielectric strength between all external connections of each mutually insulated circuit and between each external connection and the case shall be a minimum of 500V ac rms. Maximum leakage current shall not exceed one ma.

3.5.9 Magnetic effect. When positioned 5-1/2 inches from the center of a short-bar magnetic compass with a horizontal magnetic field intensity from 0.17 to 0.19, the indicator shall not cause the compass to deflect more than $1^{o}$.

3.5.10 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 equipment. The test methods shall be as specified in 4.6.14.

3.5.11 Response. The indicator shall respond to a signal equivalent to an angular change of $90^{o}$ in two seconds or less.

3.5.12 Unit torque gradient. The unit torque gradient shall be a minimum of 2,000 milligram-millimeters per angular degree of rotation for an angular displacement of $5^{o}$ of arc.

3.5.13 Effect of lead length. When the No. 20 AWG gage wire lead length between the indicator and transmitter is changed from 5 to 500 feet, the indication shall not change more than $1^{o}$ arc.

3.5.14 Power consumption. Power consumption for the indicator shall not exceed 5.0va at nominal voltage and frequency and for integral lighting shall not exceed 2.0va at 5V. (See 4.6.9)

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.



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3.7 Weight. The weight of the indicator shall not exceed 0.5 pound.

3.8 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.10 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3 Tests conditions.

4.3.1 Standard test conditions. Unless otherwise specified, the following test conditions shall apply:

a. Temperature: $25^{o} \pm 5^{o}$ C

b. Atmospheric pressure: 24 to 30.5 inches Hg

c. Relative humidity: 20 to 80 percent

d. Vibration: Indicator may be tapped or vibrated before a reading is taken

e. Position: Longitudinal axis horizontal

f. Electric power: 26 $\pm$2V 400 $\pm$5 Hz single-phase ac; 0 to 5 +0.0, -0.5Vac.

4.3.2 Test sequence. All tests shall be conducted in the order specified.



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4.3.3 Test Transmitter. The test transmitter shall be in accordance with MIL-I-7057.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of at least nine indicators representative of the production equipment. Three of the samples shall be subjected to all the test except reliability and the other six samples shall be subjected to the reliability test only. The samples shall be identified by the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. The quality conformance test shall consist of:

a. Individual tests.

b. Sampling tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Operational checks.

4.5.2 Sampling tests. Two indicators shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Insulation resistance

c. Dielectric strength

d. Leakage

e. Unit torque gradient

f. Effect of lead length

g. Lighting

h. Power consumption

i. Power variation

j. Magnetic effect

k. High temperature

l. Low temperature.



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4.5.3 Rejection and retreat. When one indicator selected from a production run fails to meet the specification, no indicator on hand or later produced shall be accepted until the extent and cause of failure have been determined. After corrections have been made all necessary tests shall be repeated.

4.5.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of indicators on hand or later produced shall not be made until it is determined that indicators meet all the requirements of the specification.

4.5.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1. Examination of products. The indicator shall be examined to determine compliance with the requirements of this specification with respect to the following:

a. Weight

b. Case and outline dimensions

c. Electrical connector

d. Exterior finish

e. Display (includes verification of readability and visibility)

f. External bonding

g. Identification of products

h. Coverglass and reflection reducing coating.

4.6.2 Operation checks. The indicator shall be connected to the applicable voltage and checked for satisfactory operation as follows:

4.6.2.1 Electrical zero. Electrical zero shall be checked in accordance with MIL-I-7057.

4.6.2.3 Scale and friction errors. The scale and friction error test shall be conducted in accordance with MIL-I-7057 with the pointer set at each dial graduation and with two indicators connected in parallel to one transmitter. The scale error shall not exceed $\pm 2.5$ angular degrees. The friction error shall not exceed $\pm 2$ angular degrees at room temperature and $+71^{o}$ C and shall not exceed $\pm 4$ angular degrees at $-54^{o}$ C.

4.6.2.4 Response. The indicator pointer shall be set in the extreme up position. The indicator shall then be energized with a transmitter equivalent to that described in this specification. The signal produced by the transmitter shall cause the indicator pointer to move smoothly to the electrical zero reference signal. The response time shall include the time taken by the pointer to settle at the electrical zero reference position. The response time shall not exceed 2.0 seconds. The test shall then be repeated with the pointer positioned in the extreme down position.



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4.6.2.5 Visual Comparison. The prime standard (see 3.4.6.1) and the indicator under test shall be placed side-by-side at eye level. The two instruments shall be viewed from a distance of two feet within a sighting cone of 30° from the normal to the center of the coverglass. The prime standard lighting circuit shall be energized with 2.85 ±0.01V and the indicator under test shall be energized with 3.00 ±0.01V. The overall brightness levels of the dial and pointer of the indicator under test shall be equal to or exceed the brightness pointer of the indicator under test shall be equal to or exceed the brightness of equivalent areas of the prime standard. The light in the specified areas shall be at least as white as the light in the corresponding areas of the prime standard. The prime standard voltage shall be increased to 3.15 ±0.01V. The brightness levels of the dial and pointer of the indicator under test shall be equal to or less than the brightness of equivalent areas of the prime standard. The light in those areas shall be no whiter than the light in the corresponding areas of the prime standard.

4.6.2.6 Fogging. The indicator shall be subjected to a fogging test in accordance with MIL-I-7057.

4.6.3 Insulation resistance. The indicator shall be subjected to an insulation resistance test in accordance with MIL-STD-202, method 302, condition B. Insulation resistance shall be a minimum of 100 megohms.

4.6.4 Dielectric strength

4.6.4.1 Preproduction. After the indicator has reached its normal operating temperature and power has been removed, the indicator shall be subjected to a preproduction dielectric strength test in accordance with MIL-STD-202, method 301. The indicator shall meet the requirements of 3.5.8.

4.6.4.2 Quality assurance. The indicator shall be subjected to a quality assurance dielectric strength test as specified in 4.6.4.1 except 375V a-c rms commercial frequency shall be applied.

4.6.5 Leakage. The indicator shall be tested for leakage by means of a mass-spectrometer type helium leak detector. The initial maximum leak rate shall not exceed 10 percent after 1,000 hours at a pressure differential of approximately 1 atmosphere.

4.6.6 Unit torque gradient. The indicator shall be subjected to a unit torque gradient test in accordance with MIL-I-7057.

4.6.7 Effect of lead length. This test shall be conducted in accordance with MIL-I-7057.

4.6.8 Lighting. The indicator shall be subjected to lighting tests in accordance with MIL-L-27160 except the life and reliability test shall be conducted in conjunction with the reliability test specified herein (4.6.18) and the dielectric strength test shall not be conducted.

4.6.9 Power consumption. At nominal voltage and frequency (26 ±2V and 400 ±5Hz), power consumption of the indicator shall not exceed the values specified in 3.5.14.

4.6.10 Power variation. The indicator shall meet the tests specified in 4.6.2.2. and 4.6.2.4 when tested at the following combinations of voltage and frequency:

a. 26V 400 Hz

b. 22V 320 Hz

c. 22V 480 Hz

d. 30V 320 Hz



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e. 30V 480 Hz.

4.6.11 Magnetic effect. The indicator shall be subjected to a magnetic effect test in accordance with MIL-I-7057 except the compass deflection shall not exceed $1^{o}$.

4.6.12 High temperature. The high temperature test shall be conducted in accordance with MIL-STD-810, method 501, procedure I. The operating temperature for step 4 shall be $71^{o}$C. At the conclusion of this test and while still at $71^{o}$C, the indicator shall meet the tests specified in 4.6.2.2 and 4.6.2.4. The indicator shall be returned to
room temperature and within a period of one hour, the indicator shall again meet the tests specified in 4.6.2.2 and 4.6.2.4.

4.6.13 Low temperature. The low temperature test shall be conducted in accordance with MIL-STD-810, method 502, procedure I. The indicator shall be subjected to a storage temperature of $-62^{o}$ C for a period of 72 hours after which the temperature shall be raised to $-54^{o}$ C and maintained for an additional 24 hours. At the conclusion of this period and while still at $-54^{o}$ C, the indicator shall pass the tests specified in 4.6.2.2 and 4.6.2.4. The indicator shall be within the specified tolerances in not more than 30 seconds. After the indicator has returned to room temperature, it shall again meet the tests specified in 4.6.2.2 and 4.6.2.4.

4.6.14 Electromagnetic interference. The indicator shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462 for class A1 equipment, methods CE03, RE02, CS01, CS04, and RS03, except as follows:

a. For method CS01, the lower frequency limit shall be 150 KHz.

b. The following shall apply in lieu of (a) test one and (b) test two of paragraph titled "Identification of Broadband and Narrowband Emissions." Broadband and narrowband emissions shall be determined as follows: any emissions shall be considered broadband if the reading of the quasi peak or peak detectors is 6db or greater above the meter readings of the field intensity (or carrier) detector functions.

4.6.15 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, method 510, procedure I. At the conclusion of this test, the indicator shall meet the test specified in 4.6.5.

4.6.16 Shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, method 516, procedure I, figure 516.1-1. The indicator need not be operated during this test. At the conclusion of this test, the indicator shall meet the test specified in 4.6.2.2 and 4.6.2.4.

4.6.17 Vibration. Prior to starting this test, the indicator shall be operated under standard ambient conditions during which it shall meet the test specified in 4.6.2.2 through 4.6.2.4. The three mutual perpendicular test axes of applied vibration, attachment of the indicator to the test fixture, and the location where the acceleration shall be monitored are shown on figure 5. The lighting circuit shall be energized with 4.5 +5, -0.0V ac during the entire test. Duration of the various portions of this test shall be as shown in table II.



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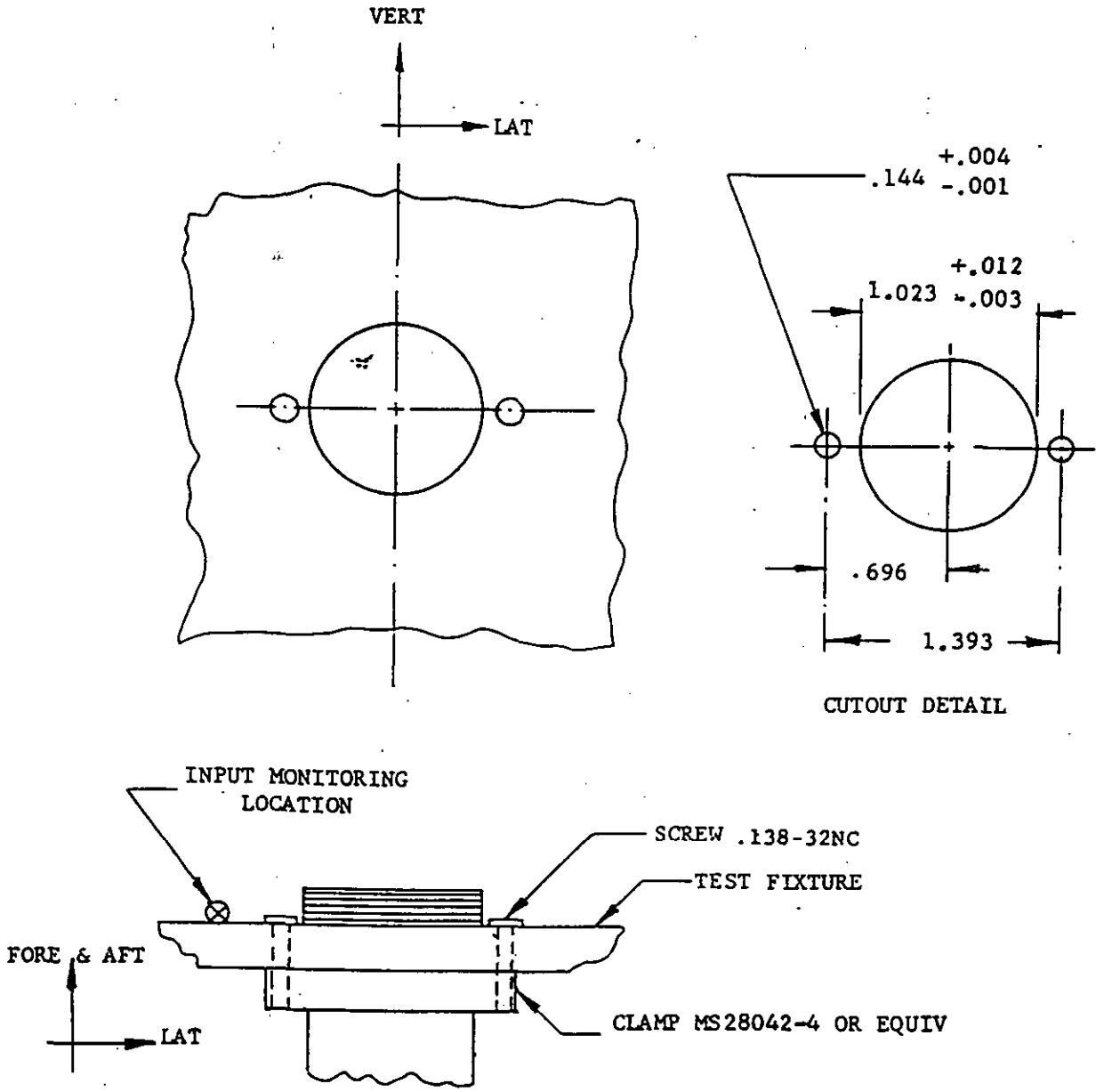



FIGURE 5. Installation for vibration test



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Table II. Sinusoidal Test Schedule

| Number of Resonances | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Total Vibration Time at Resonance* | 0 | 30 Minutes | 1 Hour | 1 -1/2 Hrs | 2 Hours |
| Cycling Time | 3 Hours | 2-1/2 Hrs | 2 Hours | 1 - 1/2 Hrs | 1 Hour |
| * 30 minutes at each axis<br><br>Duration shown are per axis | | | | | |

4.6.17.1 Sinusoidal cycling and resonance. The order in which the cycling and resonance portions of this test are conducted shall be optional. The resonant modes of the test specimen shall be determined by slowly varying the frequency between 5 and 500 Hz at an amplitude of applied vibration not exceeding that shown by the cycling test curve. When dynamically complex test fixtures or equipment is involved, low input vibration levels shall be employed to avoid unnecessary failures during the resonance search. The resonances selected for this test shall be those which are most likely to produce failure. Should any functional out-of-tolerance conditions occur during the cycling test, one of the resonance tested shall be at the frequency at which the worst out-of-tolerance condition occurred. If a change in the resonance frequency occurs during the resonance dwell, the time of occurrence shall be noted and the frequency adjusted immediately to maintain the peak resonance condition. The final resonance condition shall be recorded. During this test, the indicator shall be properly connected, energized, and the pointer set at each of the following conditions for one hour during vibration along each axis:

a. Full up

b. Full down

c. Zero.

4.6.17.2 Test procedures

4.6.17.2.1 Cycling. The indicator shall be vibrated from 5 to 500 Hz at the rate shown on figure 6 and at the amplitude shown on figure 4, curve A.

4.6.17.2.3 Monitoring. During the tests specified in 4.6.17.2.1 and 4.6.17.2.2, the indicator shall be monitored. The maximum total amplitude of pointer oscillation and variation shall not exceed $\pm 4$ angular degrees. An out-of-tolerance condition shall not be cause for rejection; however, if an out-of-tolerance condition occurs, the test specified in 4.6.17 3 shall be conducted. Following each axis of vibration test, the tests specified in 4.6.2.2 through 4.6.2.4 shall be performed.

4.6.17.3 Functional demonstration. The indicator shall operate satisfactorily when the frequency is varied between 5 and 500 Hz along each axis for period of 15 minutes at the rate shown on figure 6 and at the amplitude shown on figure 4, curve B. The maximum total amplitude of pointer oscillation and variation shall not exceed $\pm 4$ angular degrees. This test may be performed after completion of the cycling and resonance tests or at the conclusion of these tests. (This test need not be conducted if the indicator passes the cycling and resonance tests.)

4.6.18 Reliability. The six indicators shall be subjected to a reliability test which shall demonstrate a minimum acceptable MTBF of 1,500 hours to a 90 percent confidence level. Multiples of MTBF shall be in accordance with table III. No preventive maintenance shall be performed during this test. The total test time shall be computed in accordance with the following formula:



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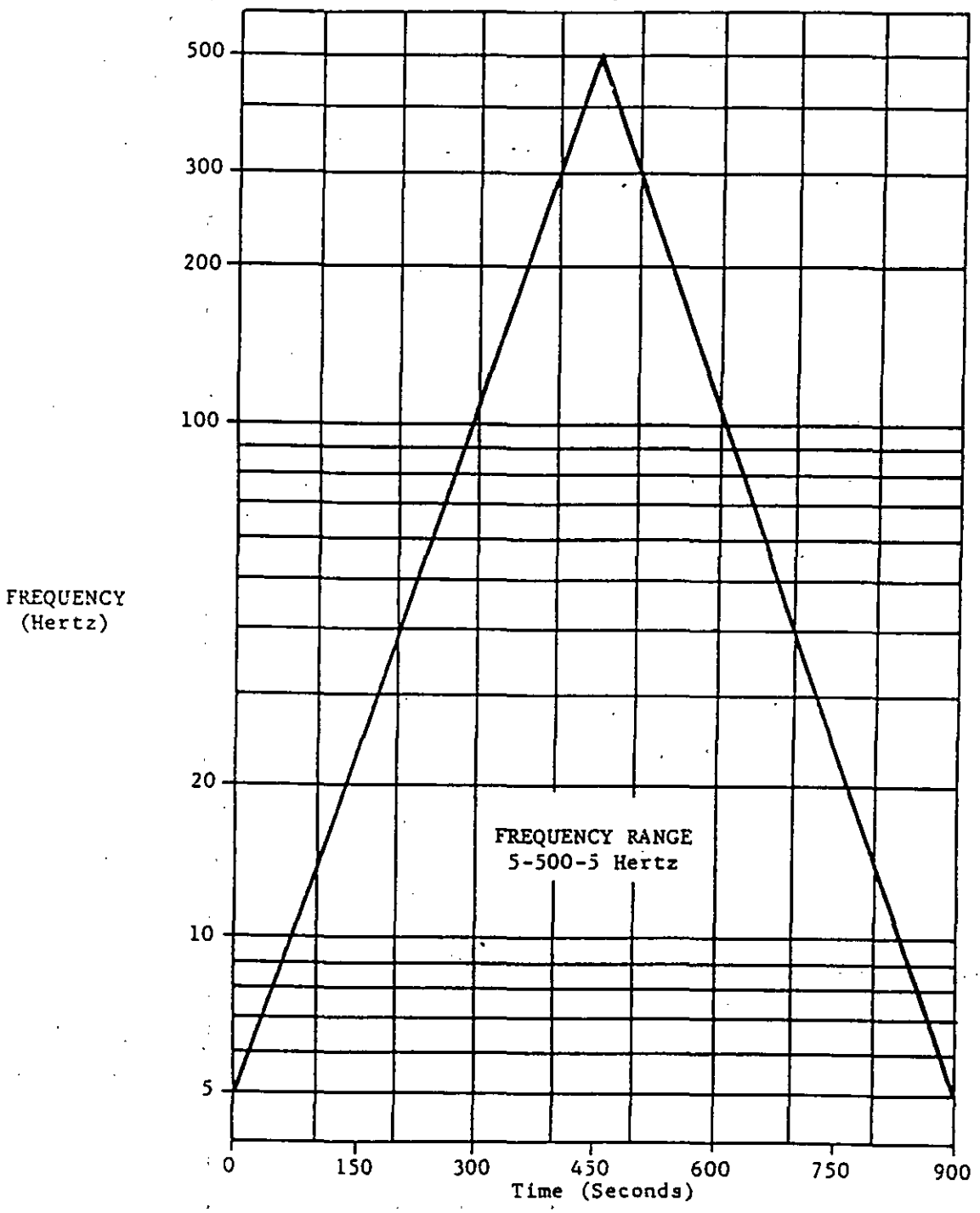



FIGURE 6. Vibration cycling test schedule



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Total test time per unit = $\frac{\text{Multiples of MTBF X MTBF X Discrimination Ration}}{\text{Number of Test Samples}}$

or

Time = $\frac{\text{MM X 1,500 X 1.5}}{6}$

4.6.18.1 Test equipment. Test equipment and fixtures shall include provisions for operation cycling, temperature cycling, and vibration.

4.6.18.2 Cycling. Throughout the reliability test, the indicators shall be cycled at a rate of 2 to 30 cycles per hour. The rate shall be changed after each 100-hour performance check.

4.6.18.3 ON-OFF Cycling. During the test specified in 4.6.18.4, the indicators shall be turned off for the first hour of the room-temperature cycle.

Table III. Reliability Multiples of MTBF

| Number of Failures | Multiples of MTBF |
|---|---|
| 0 | 4.4 |
| 1 | 5.2 |
| 2 | 6.0 |
| 3 | 6.8 |
| 4 | 7.6 |
| 5 | 8.5 |
| 6 | 9.3 |

4.6.18.4 Temperature cycling. The temperature shall be cycled as follows: Starting at room temperature, the temperature shall be increased to 71° C and stabilized for two hours, the temperature decreased to -54° C and stabilized for two hours, and the temperature then increased to room temperature and maintained for two hours. This cycle shall be repeated.

4.6.18.5 Vibration cycling. Throughout this test, the indicators shall be vibrated for a period of ten minutes during each hour at any frequency between 20 and 60 Hz at an applied acceleration of $\pm$2g.

4.6.18.6 Performance check. At each 100-hour interval and at the conclusion of the reliability test, the indicators shall be checked for scale and friction errors. During each scale and friction error check, if any two test points exceed the allowable tolerances one malfunction shall be charged against this test.

4.6.18.7 Failure criteria. For an accept decision, each indicator shall be operated within 50 hours of its total time. If an indicator fails, all testing shall be stopped and the failed unit examined to determine the cause of failure. The indicator shall be repaired and testing of the unit shall then be continued. The procuring activity shall be advised immediately of the failure, the cause of failure, and the type or repair performed. A maximum of six failures will be permitted.

4.6.18.8 Lighting. Throughout this test, the lighting circuit shall be energized except as specified in 4.6.18.3.



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4.6.18.9 Longevity. Testing of two or the reliability test samples shall be continued for a total of at least 3,750 hours on each.

4.6.18.10 Reporting. The reliability and longevity test logs shall be included in the test report specified in 4.4.2.

4.7 Inspection of packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4. Storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to ensure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.



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5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.3), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ALU-12/A indicator covered by this specification is intended for use in supersonic fighter aircraft for receiving and displaying horizontal pitch trim information.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Levels of preservation, packaging, and packing required.

6.3 Electrical insulation tests. Lighting circuit tests may be performed with the internal ground temporarily removed, when feasible; if not feasible the tests may be conducted prior to connecting the ground during assembly. Procuring activity approval must be obtained when any circuits are exempted or when filter capacitor networks and other discrete component parts on the circuit are temporarily disconnected to avoid exceeding the maximum specified design voltage.

6.4 Subject term (key words) listing. Recommended key words are:

Synchro
One Inch
Takeoff
Trim
Position Indicator

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71

Project number:
6610 - F391



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-L-83362A(USAF) | 15 DECEMBER 1994 |

3. DOCUMENT TITLE: INDICATOR, POSITION, TAKEOFF TRIM ALU-12/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | |
|---|---|---|
| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) AUTOVON *(If applicable)* | 7. DATE SUBMITTED *(YYMMDD)* |

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|---|---|
| a. NAME<br>OC-ALC/TICLA | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial (2) AUTOVON<br>336-5960 |
| c. ADDRESS *(Include Zip Code)*<br>3001 Staff Drive<br>Tinker AFB, OK 73145-5990 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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